Advance Microeconomics

Chapter #1 Game Theory

Game theory:

If I believe that my competitors are rational and act to maximize their profit, how should I take into my account their behavior, so that I might get my profit maximum.

Cooperative vs non cooperative games:

The economic theory of game might be cooperative or non cooperative.

Cooperative games	Non cooperative games
A game is said to be cooperative if players can negotiate contracts, that can allow them joint strategies.	A game is said to be non cooperative if negotiations are not possible and the firm competes to maximize their profit.
E.g if a firm decide a joint investment and agreement is signed, it will be cooperative game.	If firms make there own decisions regarding investment and competition is expected to maximize their individual profit. It will be non cooperative game.

Dominant strategies:

A dominant strategy is one that is optimal for a player, no matter what the opponent does. To explain it, we take an example on the basis of following assumptions.

Assumptions:

1. There is a duopoly setting where there are two firms say A and B.
2. They compete with each other and try to maximize their individual profit on the basis of advertisement compghain.
3. Their payoff matrix is given as follows.

		Firm B	
		Advertise	**Not Advertise**
	Advertise	10,5	15,0
Firm A			
	Not Advertise	6,8	10,2

Explanation:

The first entry in the payoff matrix shows the profit of firm A while second entry shows profit of firm B.

In case of advertisement the firm A can get profit either equal to 10 or 15 while in case of non advertisement the firm A can get profit either equal to 6 or 10. So the dominant strategy for firm A will be to advertise.

In case of advertisement the firm B can get profit either equal to 5 or 10 while in case of non advertisement the firm B can get profit either equal to 0 or 2. So the dominant strategy for firm B will also be to advertise.

Dominant strategies:

However every game has no dominant strategy for any one of the player. For this purpose we make a slight change in table 1 (All entries are same except bottom right corner). This is shown in the following table.

		Firm B	
		Advertise	**Not Advertise**
	Advertise	10,5	15,0
Firm A			
	Not Advertise	6,8	20,2

The given table shows that firm A has no dominant strategy. Firm B has dominant strategy (It earns profit equal either equal to 5 or 8 if it

advertises). Thus in this case firm A will depend upon the decisions of firm B.
Firm A has two options either to get profit equal to 10 or 6. So it will surely opt 10.
Thus we conclude that both the firms will advertise.

Nash equilibrium:

This equilibrium concept was presented by John Nash in 1951. There is a fundamental difference between dominant strategy and Nash equilibrium which is as follows.

Dominant strategy:

1. I am doing the best, I can no matter what you do.
2. You are doing best, you can no matter what I do.

Nash equilibrium:

1. I am doing the best, I can given what you are doing.
2. You are doing the best, you can given what I am doing.

Example:

Consider a product choice problem. Two cereal producing firms guess that two new variations can be launched successful. They are the sweet and crispy cereal. Each firm has resources to launch only one variation. The payoff matrix is given in the following table.

		Firm 2	
		Crispy	**Sweet**
	Crispy	-5,-5	10,10
Firm 1			
	Sweet	10,10	-5,-5

Firm 2 decides to produce sweet cereal and this information is attained by firm 1. Firm 1 will produce crispy cereal depending upon the behaviour of firm 2. Thus the equilibrium will be found in top right corner where both firm will earn the profit equal 10 each.

The Nash equilibrium can happen at more than one places. Let we assume that firm 2 decides to produce crispy cereal. So the firm 1 depending upon the behaviour of firm 2 will surely produce sweet cereal. Thus equilibrium will be found in bottom left corner where both firm will earn profit equal to 10.

Repeated games:

We know about the Prisoners Dilemma which shows the static situation. Now the question is how firms can come out of dilemma to maximize its individual profit / output. It means there must be repeated games.
To understand it, look at the following payoff matrix.

		Firm 2	
		Low price	**High price**
	Low price	10,10	100,-50
Firm 1			
	High price	-50,100	50,50

The matrix shows that if you and your competitor charge the high price, both of you will get marginal profit equal to 50 for each.

You are afraid that if you charge high price and your competitor charges low price, then you will face loss. Now question is that how to minimize your loss, obviously that will be tit for tat policy which is, if your competitor charges low price then you have to charge low price and if your competitor charges high price then you have to charge high price in order to minimize your loss.

Threats, commitments and credibility:

- **Advantage of moving first:**

If there is advantage in producing one of the two commodities, the firm which moves first will reap that advantage. For explanation, look at the following payoff matrix.

		Firm 2	
		Low price	**High price**
	Low price	-5,-5	10,20
Firm 1			
	High price	20,10	-5,-5

The matrix shows that there is advantage in producing the sweet cereal. If the firm1 has better resources and more time than firm2, it will choose to produce sweet cereal. Thus equilibrium will be found in lower left corner. Firm 1 will get profit equal to 20 and firm 2 will get profit equal to 10.
This is the case when the firm1 has more time to start the business.
The other case is if both of firms have equal time and similar circumstances, In this case commit will be better strategy. Anyone of the firm will commit to produce sweet cereal knowing that its opponent will come under its threat and produce the crispy cereal. For commitment the firm may use the advertisement. Furthermore, it may spread rumors in the market. Commitment might be empty threat but it will force the other firm to produce the crispy cereal. Thus the firm committing threat will reap the advantage.

Empty threats:

These are the threats which have no weight age in practical life.

Explanation:

To understand empty threat, we assume

1. There are two firms which produce personal computers.
2. Firm1 produces PCs that can be used as word processors to do other tasks.
3. There profit is shown in following payoff matrix.

		Firm 2	
		High price	**Low price**
	High price	10,5	15,0
Firm 1			
	Low price	6,8	10,2

The matrix shows that charging high price is dominant strategy for firm 1 while charging low price is the dominant strategy for firm 2. If firm 1 charge high price and firm 2 charge low price, the equilibrium will be found in upper right corner which is not ideal position for firm 2. The ideal situation for firm 1 is upper left corner. For this purpose it desire firm 2 should also charge the high price. For this purpose it will force firm 2 to charge high price otherwise it will charge low price and equilibrium will be found in lower right corner where both firms will have to be significantly. But loss to firm 1 is greater than firm 2, so the threat is not credible.

Commitment and credibility:

Let us assume that there are two firms. Firm A produces engines while firm B produces cars. It is more profitable for firm B to purchase engines from firm A. Firm B will decide to produce cars (small or big). Firm A will follow firm B. Such games are called sequential games. There profit in million dollars is shown in the following payoff matrix.

		Firm B	
		Small cars	**Big cars**
	Small engines	3,6	3,0
Firm A			
	Big engines	1,1	8,3

The matrix shows that the dominant strategy for firm A will be lower right corner where it earns profit equal to $8 million while firm B will earn profit equal to $3 million.
Firm A can induce firm B to produce big cars and it is more suitable for firm B to purchase engines from firm A.
Firm A will lose some profit in its payoff matrix to make its threat more credible. This is shown in the following payoff matrix.

		Firm B	
		Small cars	**Big cars**
	Small engines	0,6	0,0
Firm A			
	Big engines	1,1	8,3

Firm A shuts down producing small engines or destroys the available stock of small engines which firm B realizes. Thus the threat becomes more credible. Firm A will produce big engines and firm B will also produces big cars and equilibrium will be found in lower right corner.

<u>Entry Deterrence:</u>

***<u>Definition</u>*:**

When an existing firm discourage new firm to enter in the market, we call it entry deterrance.

<u>Explanation:</u>

To explain it through game, we assume the following:

1. There is a firm say A which is enjoying monopoly power in the market.
2. There is a potential competitor say B wants to enter in the market.
3. The potential competitor has sunk cost equal to $40 million.

The whole process of possible profit is shown in the following payoff matrix.

	Firm B

		Enter	Stay out
	High price	50,10	100,0
Firm A			
	Low price	30,-10	40,0

If firm B decides to enter and firm B charges high price, the equilibrium will be found in top left corner. (Firm A will earn profit equal to $50 million and firm B will earn profit equal to $10 million.
Firm A's dominant strategy is found in top right corner (where it wishes firm B to stay out of the game). Firm A threats firm B to stay out of the game otherwise it will charge low price and firm B will face loss equal to $10 million.
The equilibrium will be found in lower left corner. Firm B realizes that in lower left corner if firm B faces loss equal to $20 million, firm A will also face loss equal to $20 million. The threat will not be credible.
To make the threat more credible, we make a slight change in the payoff matrix.

		Firm B	
		Enter	Stay out
	High price	20,10	70,0
Firm A			
	Low price	30,-10	40,0

Firm A desires upper right corner while firm B desires upper left corner. If firm B insists to enter, firm A will charge low price and get more profit than upper left corner. (Desired equilibrium of firm A). The threat is credible and firm B will stay out.

Bargaining Strategy:

Bargaining strategy is enforceable agreement.

***Explanation*:**

We consider the following payoff matrix.

		Firm B	
		Produce A	**Produce B**
	Produce A	40,5	50,50
Firm 1			
	Produce B	60,40	5,45

The matrix shows that acceptable situation is possible either at lower left corner or top right corner, the enforceable agreement is not possible at lower left corner. Firm 2 will not agree because in this equilibrium firm is having profit equal to $60 million while firm 2 is having profit equal to $40 million.
The enforceable agreement is possible at upper right corner where both of the firms are earning profit equal to $50 million.

Chapter # 2 Exchange

Exchange:

Exchange brings economic efficiency. Here we study efficient allocation of two goods between two consumer. Initially the concept of exchange is developed on the basis of concept of trade.

The advantages of trade:

The trade is beneficial between two consumers when it brings efficient allocation of goods they are having.
Let we have the following assumptions.

1. There are two people having two goods (Ahmed and Ali having food and cloth).
2. Both people have complete information about their preferences.
3. Transaction cost is 0.

The advantage of trade is shown in the following table.

Individuals	Initial allocation	Trade	Final allocation

Ahmed	7F, 1C	-1F, +1C	6F, 2C
Ali	3F, 5C	-1C, +1F	4F, 4C

The table shows that Ahmed has 7 units of food and 1 unit of cloth, while Ali has 3 units of food and 5 units of cloth in initial allocation.

According to the law of DMU, Ahmed desires cloth and Ali desires food. Thus they will trade in both commodities with each other. Ahmed will leave 1 unit of food and Ali will leave 1 unit of cloth.

In the final allocation, both Ahmed and Ali will have efficient allocation.

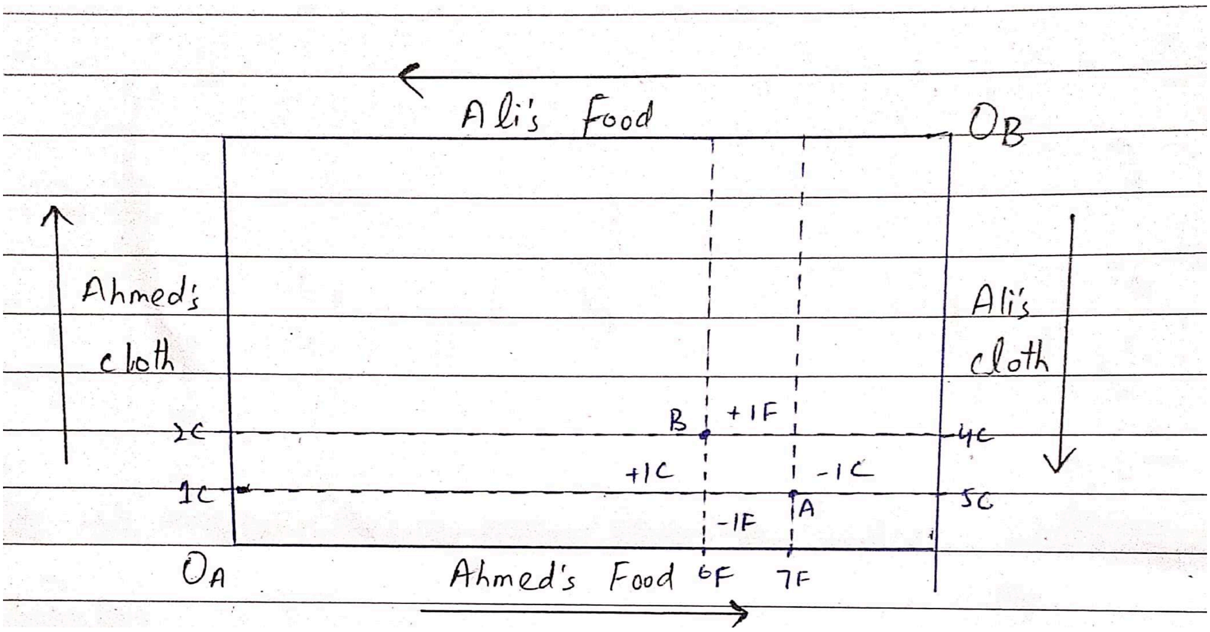
<u>The Edgeworth Box diagram:</u>

The Edgeworth box is named after political economist F.Y Edgeworth who suggested the use of edgeworth box in 1881 in his book " An essay on the application of mathematics and statistics in the moral sciences ".

There might be questions regarding trade.

1. Which of the traders will allocate goods efficiently among consumers.
2. How much consumer will be better off trade?

We can answer these questions by assuming two persons (Ahmed and Ali) and two goods (Food and Cloth) by using Edgeworth box diagram which is given below.



The diagram shows that initial allocation is found at point A while final allocation is found at point B.

Efficient Allocation:

Now we have to see whether two goods (food and cloth) are efficiently allocated between Ahmed and Ali at point B.
The efficient allocation of goods depends on MRS. The goods (food and cloth) will be efficiently allocated between Ahmed and Ali where

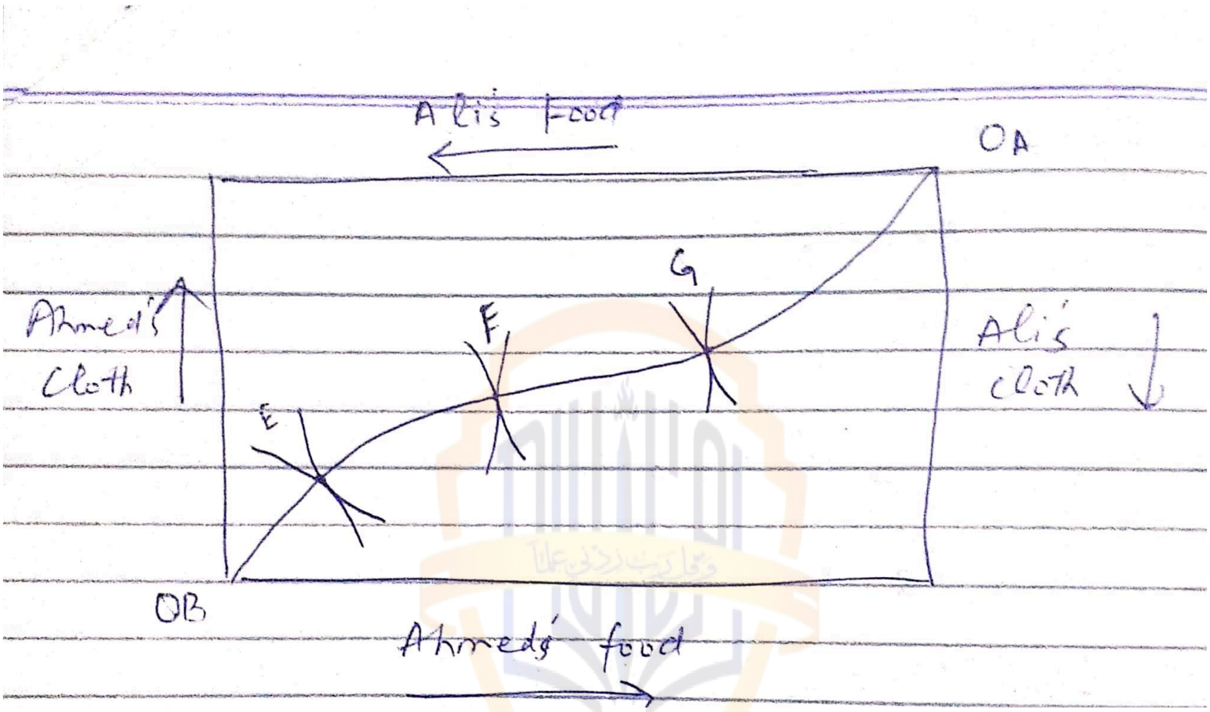
MRS(ali) = MRS(ahmed)

MRS depends on ICs so we have to draw ICs of both consumers to find efficient allocation of both goods (food and cloth). Earlier we decided point after trade but we are to see whether we see that at point A, Ahmed's ICB and Ali's ICA are intersecting each other, so both goods are not efficiently distributed here. After trade Ali sacrifices 1 unit of cloth to get an additional unit of food. Similarly Ahmed sacrifices 1 unit of food to get an additional unit of cloth. We reach at point B. Again at point B, both of goods are not efficiently distributed between two consumers because ICs of both Ahmed and Ali are intersecting at point B. Again some units are traded and we reach at point C. The goods are efficiently allocated because at point C

MRS(ali) = MRS(ahmed)

The Contract Curve:

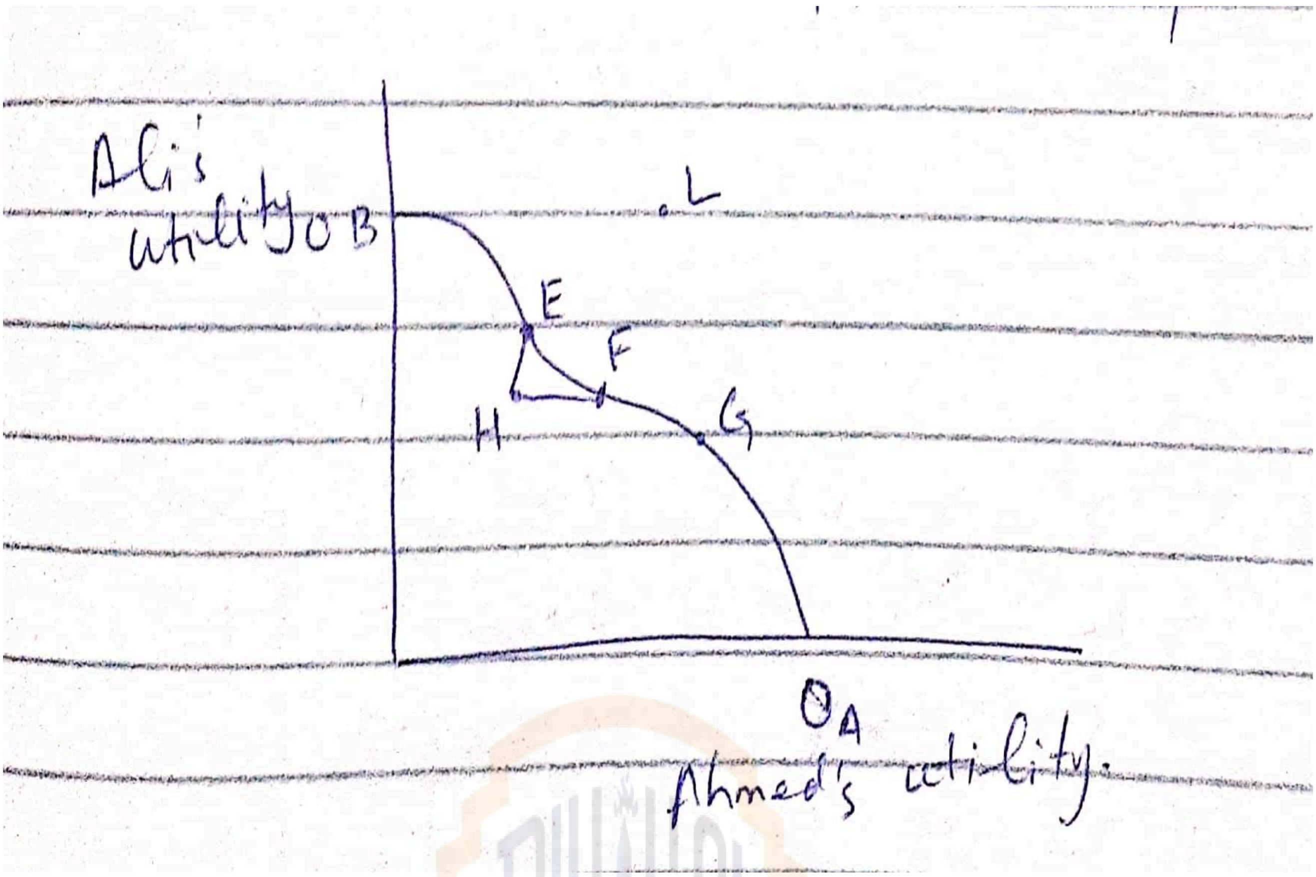
The contract curve contains all allocation for which consumer's indifference curves are tangent. Every point on the contract curve is efficient because one person cannot be made better off without making the other person worse off.
These allocations are some times called Pareto-efficient allocations, after the Italian economist vilf-redo Pareto who developed the concept of efficient exchange.
This is shown in the following figure.



In box point E F and G are most efficient points because they are found on the same contract curve. Any other point in the box than the contract curve will not be Pareto efficient point because by doing so, if one person is better off the other will surely be worse off.

Utility possibility frontier:

The utility possibility frontier shows the levels of satisfaction that each of the two people achieve when they have traded to an efficient outcome on the contract curve.
This is shown in the following figure.



The figure has been constructed with the help of Edgeworth box diagram. Every movement towards right shows increase in Ahmed's utility while every upward movement shows increase in Ali's utility. The utility possibility frontier having points E,F and G shows efficient allocation of two commodities between Ahmed and Ali. The point H is inefficient because when we move from H to F, there is increase in utility of Ali without decreasing the utility of of Ahmed. Similarly if we move from H to F, there is increase in utility of Ahmed without decreasing utility of Ali. The point L is away from utility frontier. Only point E,F and G show efficient allocation. (This is utility possibility frontier).

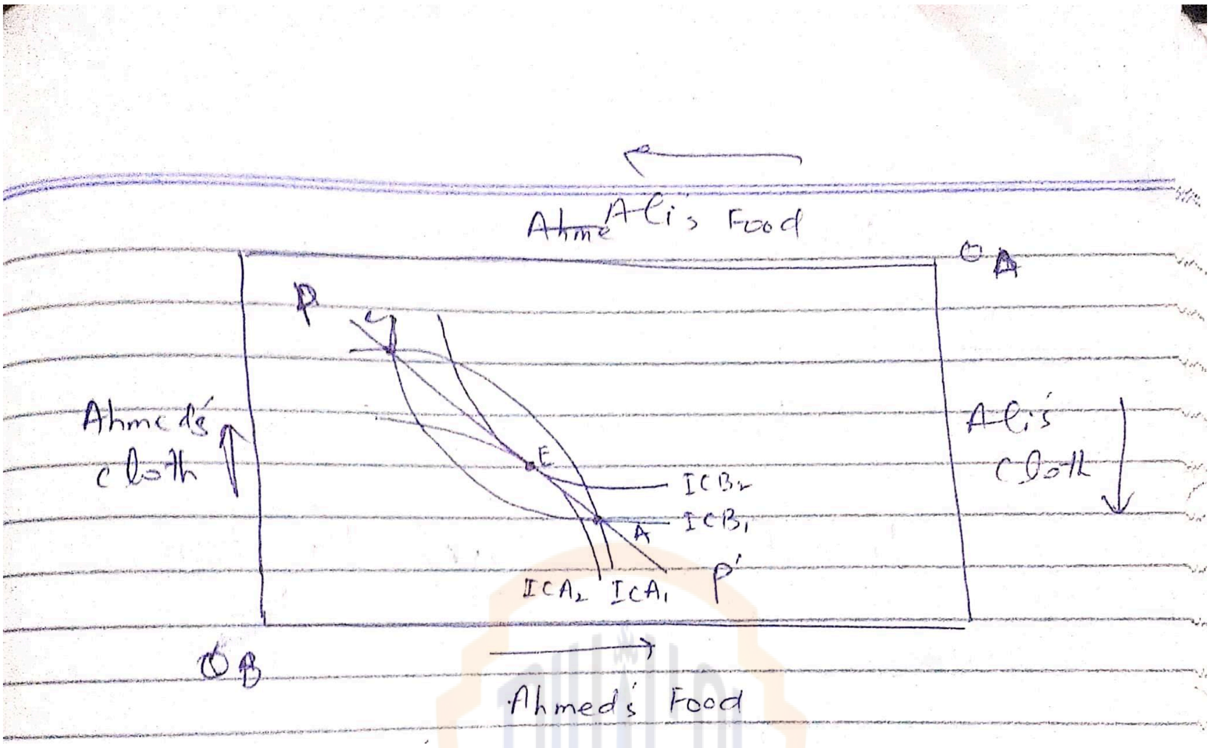
Consumer Equilibrium in competitive Market:

In competitive market there are many buyers and sellers, their exchange depends on their terms of exchange. In competitive market the prices of two goods determine the terms of exchange. We assume that Ahmed and Ali each represents an average consumer from a large group of consumers.

The equilibrium will be found where Ali's IC and Ahmed's IC are tangent on the same point of price line (Budget line).

$MRS_{(Ali)} = P_x / P_y = MRS_{(Ahmed)}$

To show this situation we draw the following Edgeworth box diagram.



In the given diagram pp' shows price line of two consumer. ICA1 and ICA2 are ICs of Ali and ICB1 and ICB2 are ICs of Ahmed.

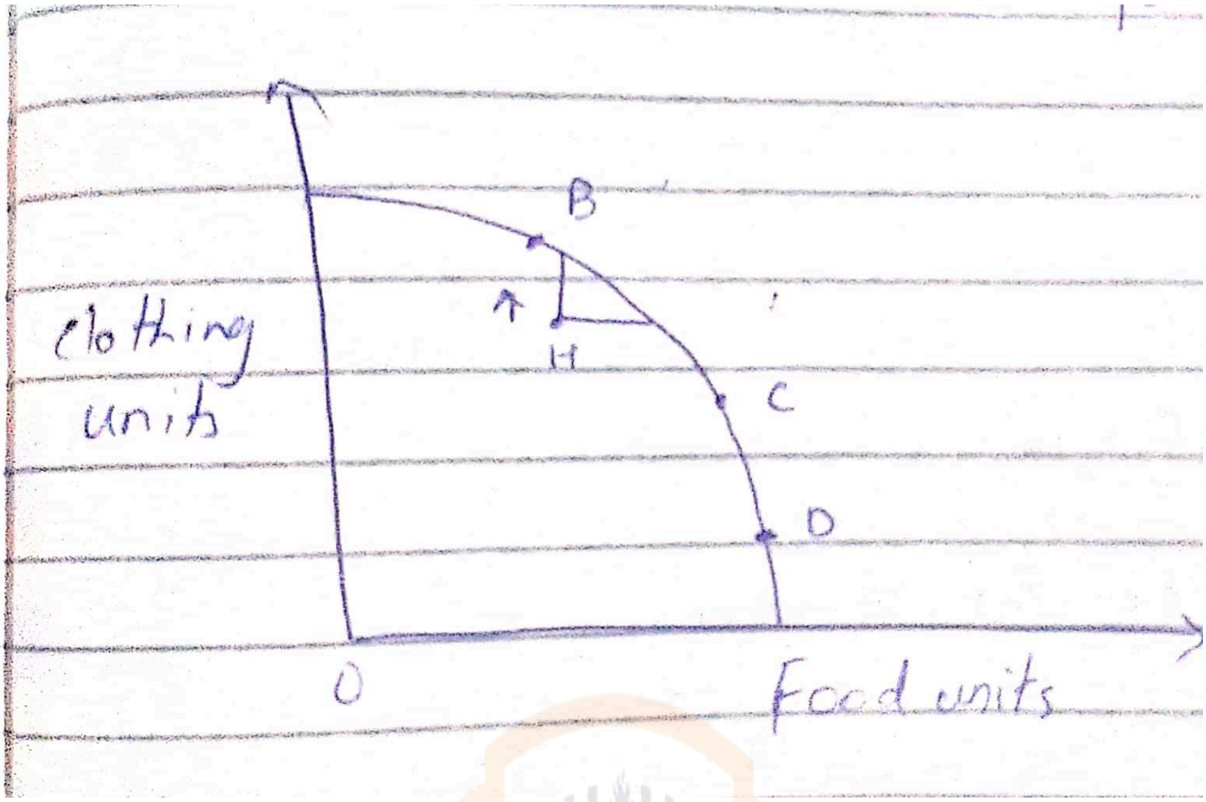
Let we start our discussion about at point A or C. At both of these points the ICs of both consumers are intersecting each other, so the condition of consumer equilibrium in competitive market is not filled. Both consumer moved towards point E. At point E the condition of consumer equilibrium in competitive market is filled because at this point,

$MRS_{(Ali)} = P_x / P_y = MRS_{(Ahmed)}$

Production possibility frontier:

The production possibility frontier, shows the various combinations of food and clothing, that can be produced with fixed inputs (labor and capital). It shows all efficient combinations of outputs.

This is shown in the following figure.



It is a concaved curve due to MRT. All the combinations like A,B,C,D on frontier are Pareto efficient points. All the points below frontier are inefficient point, for example point H. If we move vertically from point H to B we whould be able to get extra units of clothing without leaving any unit of food. It shows increase in efficiency of output.
Again if we move horizontally from point H to C, we whould be able to get extra units of food without leaving any unit of clothing. It shows increase in efficiency of output.
So it is concluded that all points on production possibility curve like B,C and D are Pareto efficient points.

Efficiency in output market:

In a perfectly competitive market, all consumers allocate their budget in such a way that their marginal rate of substitution becomes equal to the price ratio of the goods they are consuming. It means that $MRS = P_F / P_c$

At the same time prices represent marginal cost.

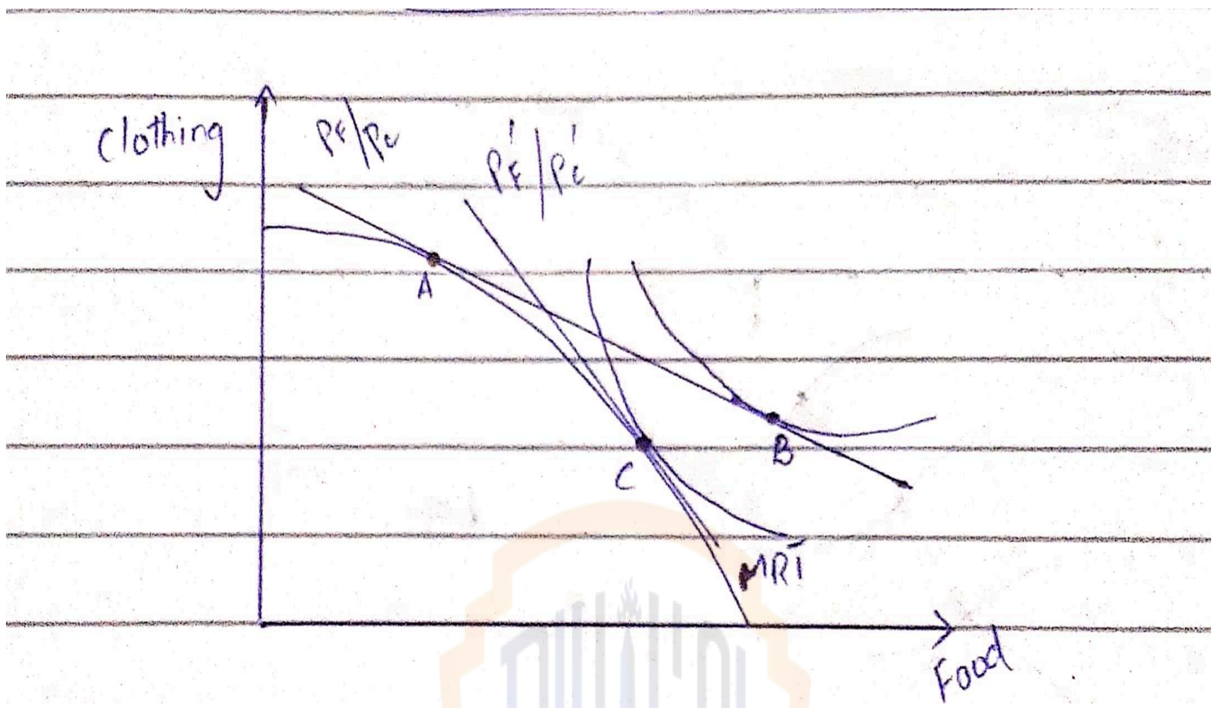
$Pf = MCf$

$Pc = MCc$

Thus $MRS = Pf / Pc = MCf / MCc$

As we know that $MCf / MCc = MRT$

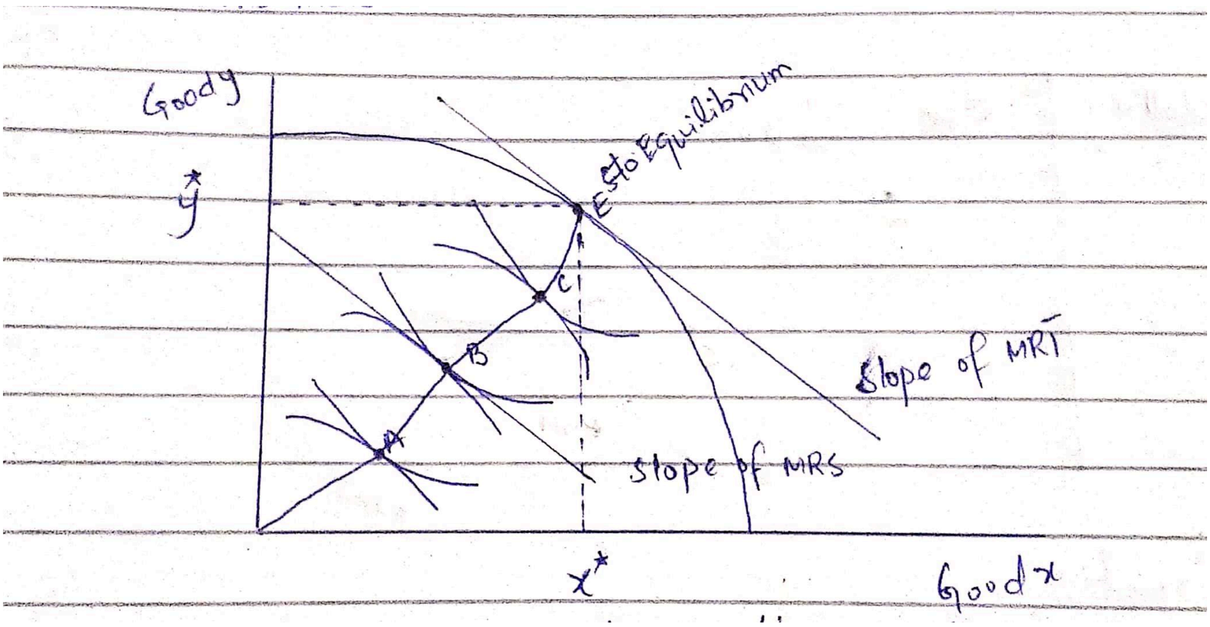
Then
$MRS = Pf / Pc = MCf / MCc = MRT$



The given diagram shows that there is point A on budget line Pf / Pc. At point A, MRT = Pf / Pc but it is not equal to MRS because MRS is found at point B on the same budget line. So we draw another budget line Pf ' / Pc' and at this budget line we found equilibrium point C. At point C both MRT and MRS are equal to Pf ' / Pc', as shown in the figure.

<u>Pareto Efficiency:</u>

Pareto efficiency will be found where MRS = MRT for goods x and y between two consumers.



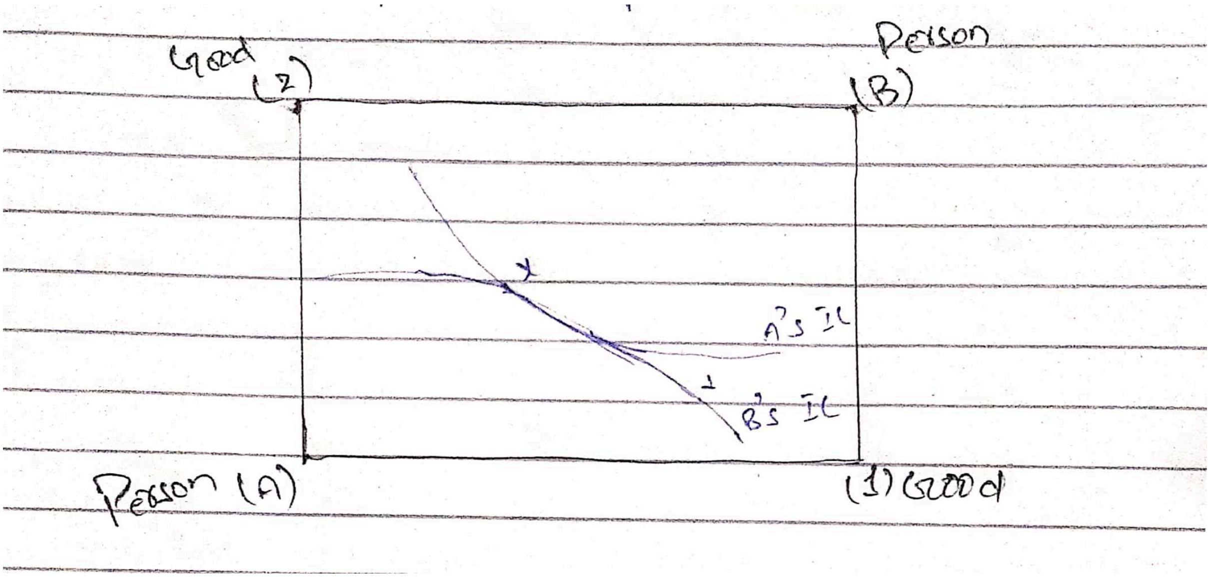
The given diagram shows that there are three points A, B and C called pareto efficient points, any other point in the diagram than A, B and C will not be pareto efficient point. Similarly point E on utility possibility frontier is also called pareto efficient point. On point E we found equilibrium points x* and y*. At point E slope of MRS = slope of MRT.

First welfare theorem:

The first welfare theorem is in fact a mathematical restatement of Adam Smith's famous " invisible hand " result. It states

A competitive equilibrium is pareto optimal.

According to this theorem, a person A chooses the point X on B's indifference curve through the highest utility. Such a point must be pareto efficient. This is shown in the following figure.



In the diagram the person A will choose the point X rather than V.

Implications of first welfare theorem:

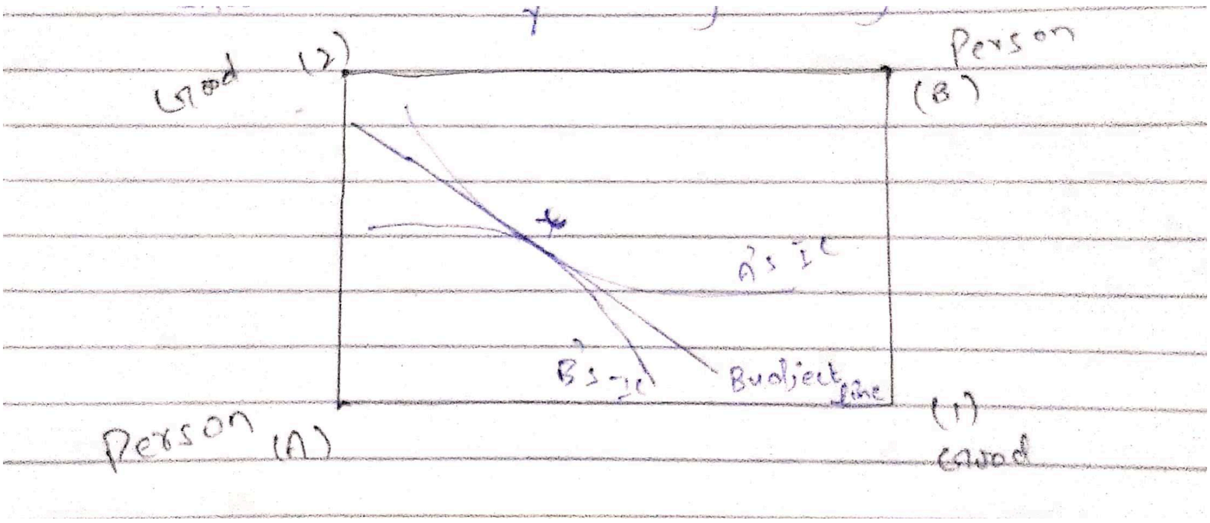
1. We have demonstrated this theorem in the simple Edgeworth box case but it is true for more complex models where there are many consumers and many goods.
2. One agent does not care about another consumption. We can say that there is consumption externalities.
3. The agents will use their market power to improve their own position.
4. It is helpful to ensure pareto efficient point.

Second welfare theorem:

According to this theorem, when preferences are convex, a pareto efficient allocation is an equilibrium for set of prices.

Explanation:

The concept can be explained by geometric arguments. If consumer A and consumer B have convex preferences, we can draw a straight line between the two sets of preferred bundles that separates one from the other. The slope of the line gives relative prices.
An endowment that puts the two agents on this line will lead to find market equilibrium is the pareto efficient. The whole process is shown in the following figure.



Endowment x is pareto efficient.

Economic implications of 2nd welfare theorem:

1. This theorem emphasizes that under certain conditions (relative prices) pareto efficient allocation can be achieved as a competitive equilibrium.
2. Prices play two rules in the market system. One is the allocative and the other is distributive. These two rules can be separated.
3. Taxing labor is considered discretionary tax. The second welfare theorem proves it how discretionary tax.
4. Charging low price from small user of electricity is according to the concept of relative price. Which is given second welfare theorem.

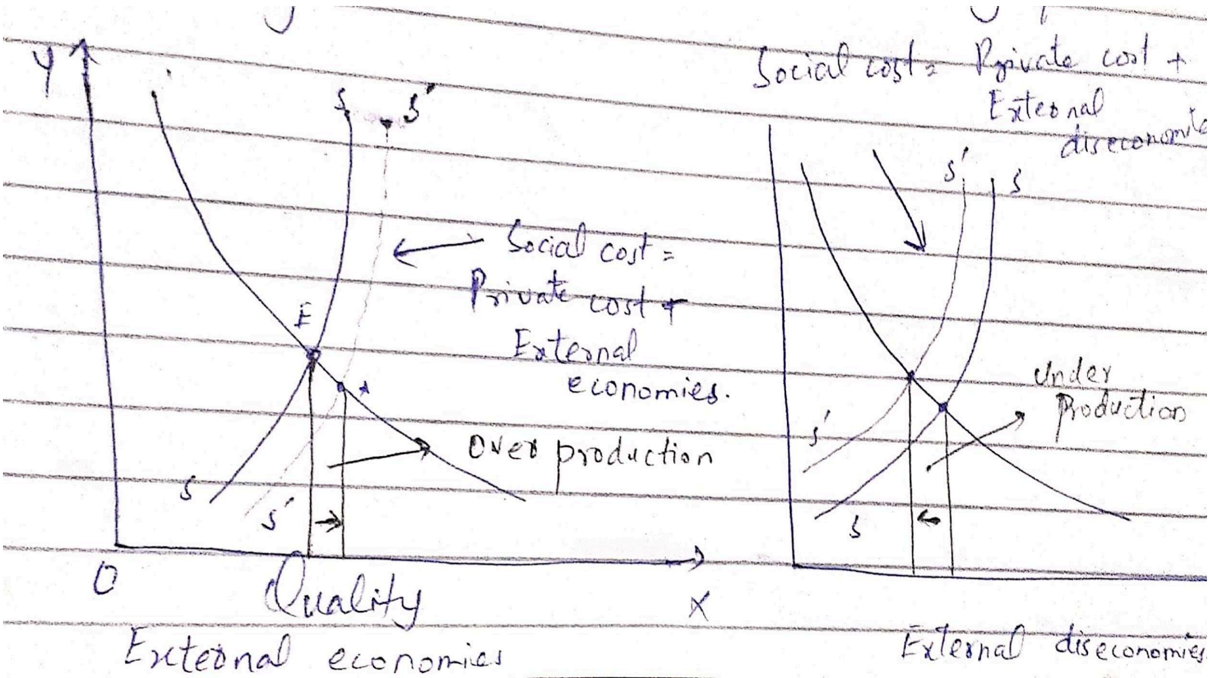
How externalities cause market failure:

The question rises that how externalities can lead to misallocation of resources. When there are externalities in the market, private cost diverges social cost which lead to misallocation of resources and disturb pareto optimality.

When there are external eonomies, (beneficial effect) of production, private marginal cost will be greater than social marginal cost.

When there are external diseconomies (loss effect) of production, private marginal cost will be lesser than social marginal cost.

This is explained in the following panel of diagram.

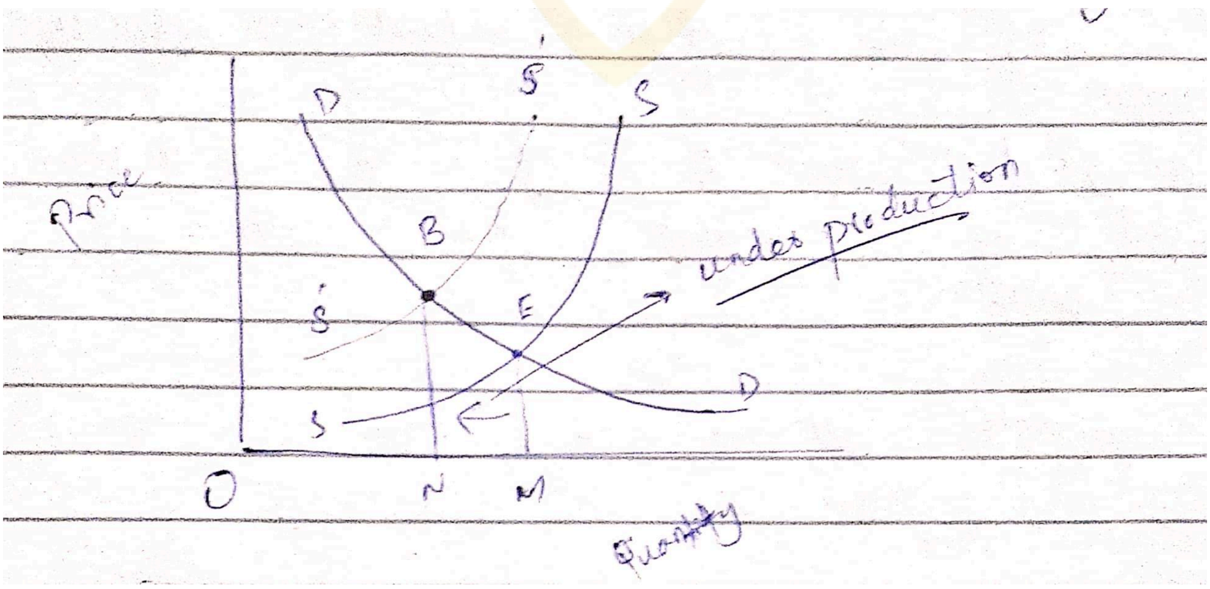


<u>**Free riders problem and public good:**</u>

The public good has characteristics of non- excludibility. The producer of good has inability to exclude anyone from use of that good. This introduces the concept of free riding.

Free riding leads to market failure and non- existence of pareto optimality. If a private producer is asked to produce public good. He will not agree to produce that good or he will produce it in less quantity than required.

This is shown in the following diagram.



The diagram shows that equilibrium should be at point E but the producer will produce the less quantity than required. This pushes supply curve SS back to S'S' and there is under production equal to MN. This leads to market failure and non existence of pareto optimality.